

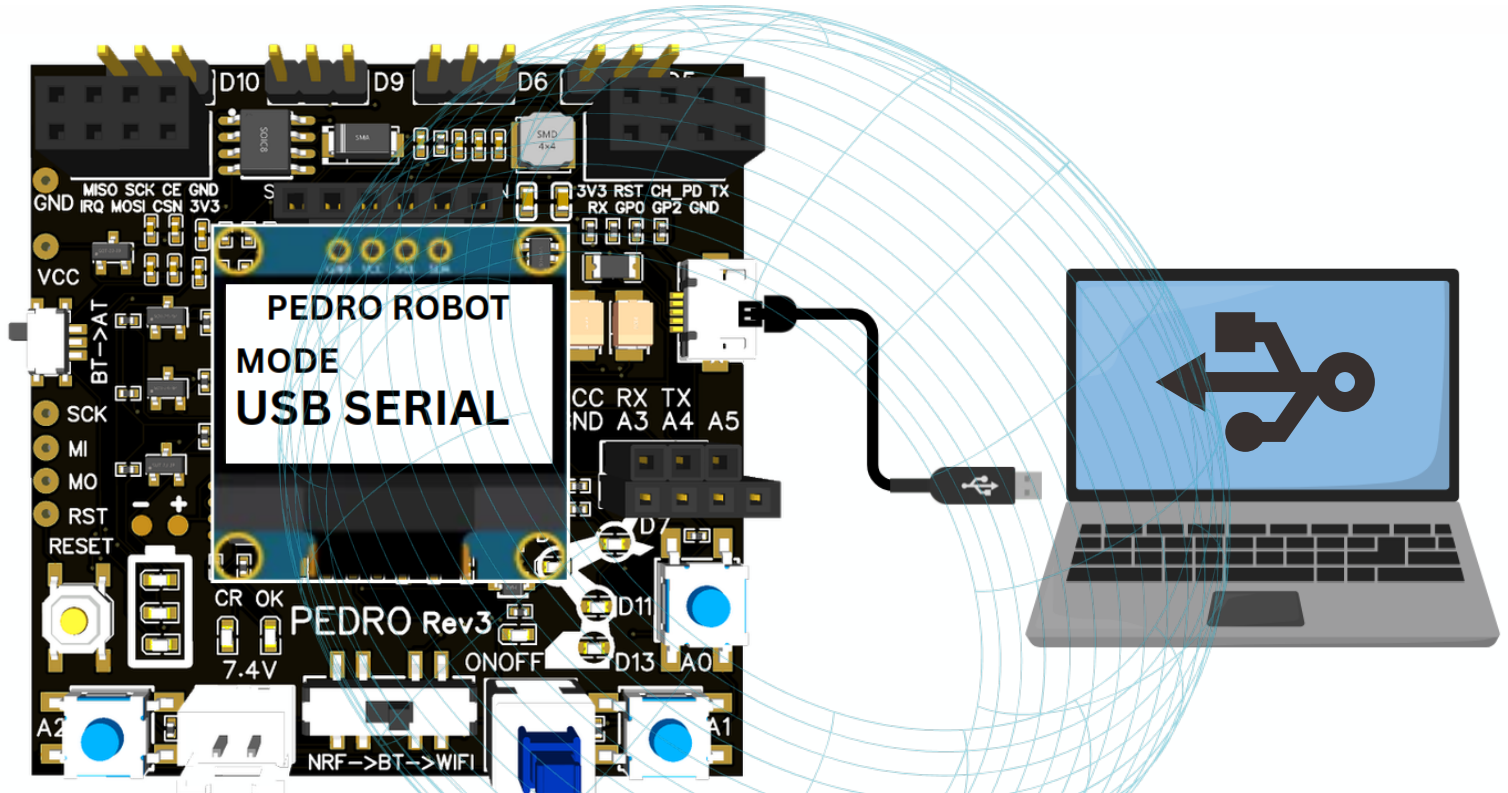
BEGINNER

pedro



STEM Program

Lesson #6: USB Serial Mode



Complete this lesson
to earn your Pedro
Badge



Add it to your Pedro
Passport to track
your STEM journey

YOUR MISSION: Learn how robots exchange data with computers using USB Serial communication.

Learning Goals

In this lesson, you will discover how to control Pedro directly from a computer using USB Serial communication. You will learn how a computer exchanges data with a robot through an USB cable and how digital commands are translated into robotic movement. By the end of this lesson, students will be able to:

- Understand USB Serial communication
- Explain how computers communicate with robots
- Control servomotors from a web application
- Understand serial data transmission

At the end of this lesson, you will unlock your Pedro Badge. 🏆 the Pedro USB Serial Badge



Teacher Notes

- No prior robotics experience required.
- Duration: 60–90 minutes
- Difficulty: ● Beginner
- Required Materials
 - **1 Assembled** Pedro robot **per student or per group** (*recommended: groups of 2 students*)
 - **OLED Screen 128X64** and **USB cable**
 - **1 computer** with the Arduino IDE installed



Think: The **brain icon** encourages students to make careful observations, share their ideas, and engage in discussion by explaining and justifying their answers.



Learn: The **light bulb icon** indicates important concepts that students should read carefully. Students are encouraged to ask questions and discuss these topics about the robotics concepts presented in the lesson.



Make: The **pencil icon** indicates that students must complete the activity on the page. These activities are designed to reinforce the concepts learned through experimentation, and problem-solving.



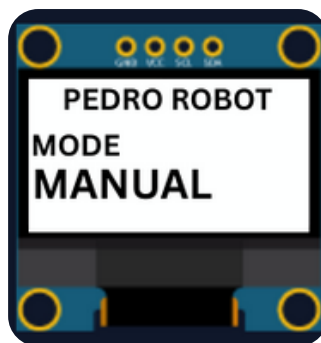
Board Configuration

(Skip this activity if the Pedro program is already uploaded)

The **Pedro Program** is the core software that powers and controls the robot. It acts as the robot's "brain," managing all behaviors and interactions. Through this single program, students can access and explore multiple control modes, allowing them to understand how a robot receives instructions, processes them, and translates them into physical actions.

1. **Insert** the Oled Screen into the Pedro board
2. **Connect** the board to your computer with the Micro USB cable.
3. Open **Arduino IDE**
4. **Install the required libraries** from the Library Manager:
 - Tools → Manage Libraries → search **PedroRobot** → Install
 - Tools → Manage Libraries → search **U8glib** → Install
 - Tools → Manage Libraries → search **RF24** → Install
5. Select the port that appear when you connect Pedro robot: **Tools → Port**
6. Select the board: **Tools → Board → Arduino AVR Boards → Arduino Micro**
7. Open the PedroRobot program: **File → Examples → PedroRobot → Pedro**
8. **Compile and upload** the program to your Pedro board.

✓ If everything is correct, Pedro's OLED screen will display **MANUAL MODE**.



Press **A0** and observe what happens. Each time you press **A0**, Pedro selects the next servo. The LED corresponding to the selected servo lights up, indicating which part of the robot you are controlling. The mapping between the LEDs and the servos is shown below:

- LED D13 → servo D5 (Base)
- LED D11 → servo D6 (Shoulder)
- LED D8 → servo D9 (Elbow)
- LED D7 → servo D10 (Gripper)



USB Serial communication

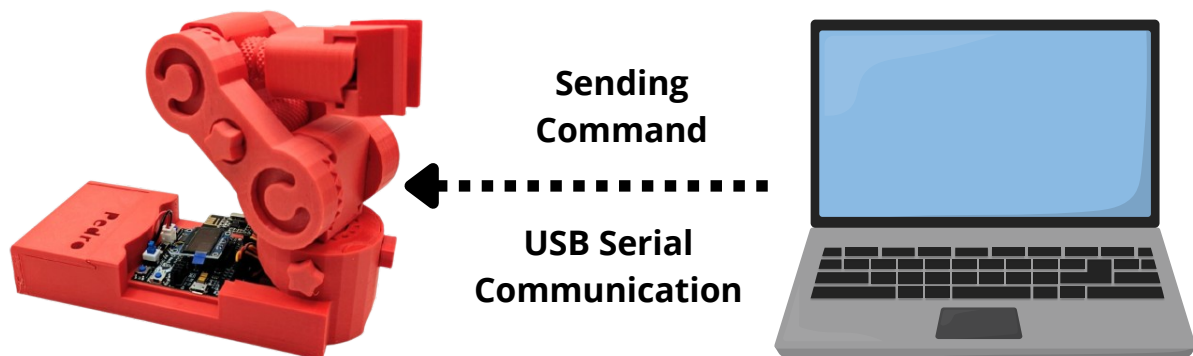
Before wireless communication existed, robots were commonly controlled through wired serial communication. Even today, USB Serial communication is still essential in:

- robotics
- industrial automation
- spacecraft systems
- embedded engineering

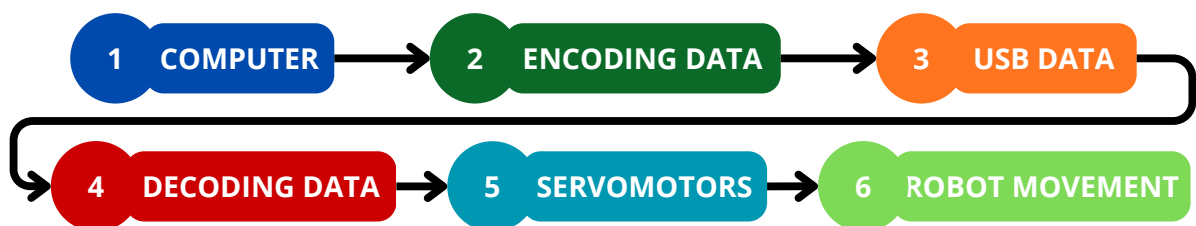
USB Serial communication allows a computer to:

- send commands
- receive data
- monitor sensors
- control robotic systems

Pedro uses this same engineering principle. USB Serial communication works like a digital conversation between the computer and the robot. The computer sends digital commands through the USB cable. Pedro receives the commands and executes actions.

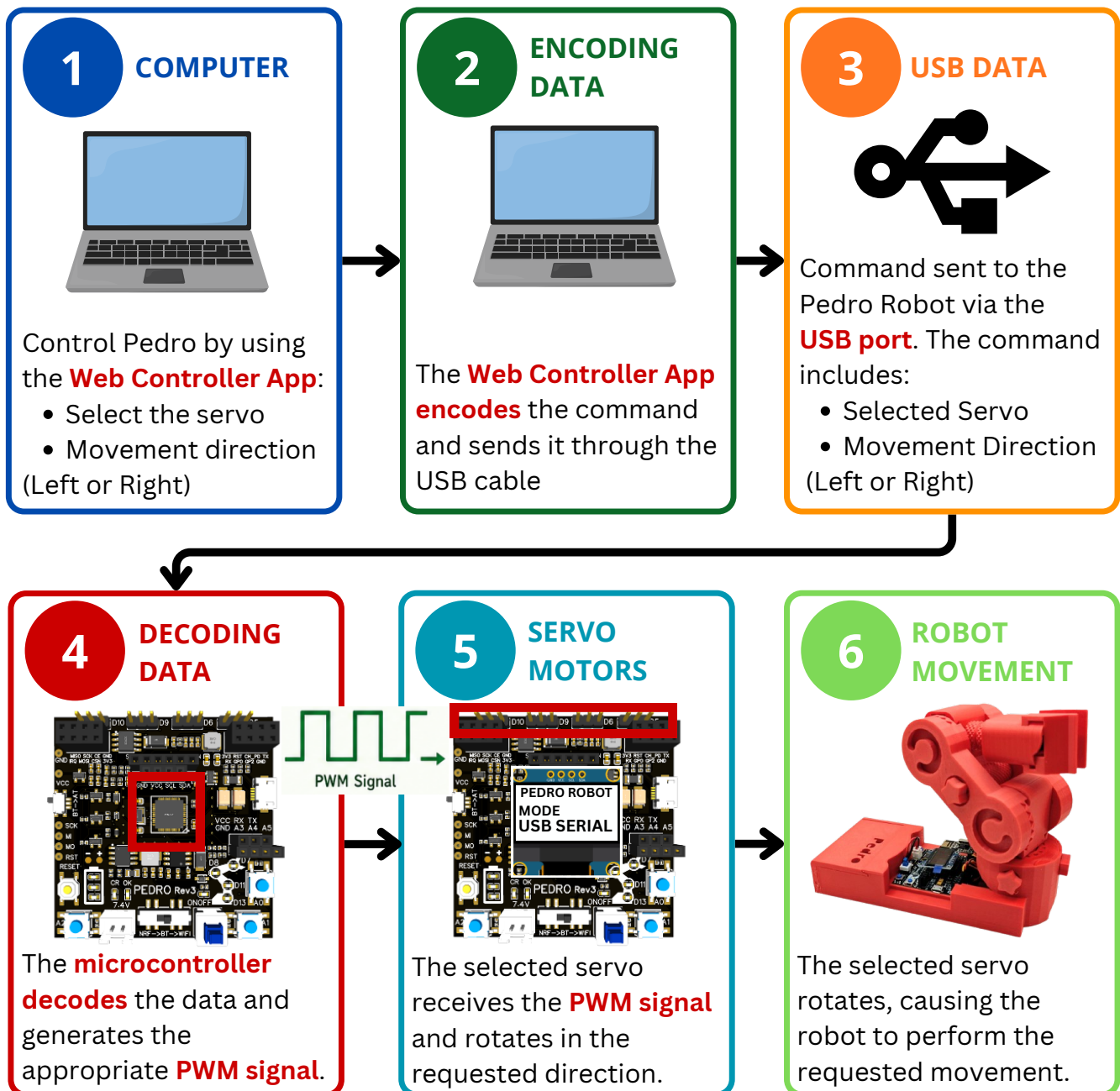


The USB Serial communication chain is:



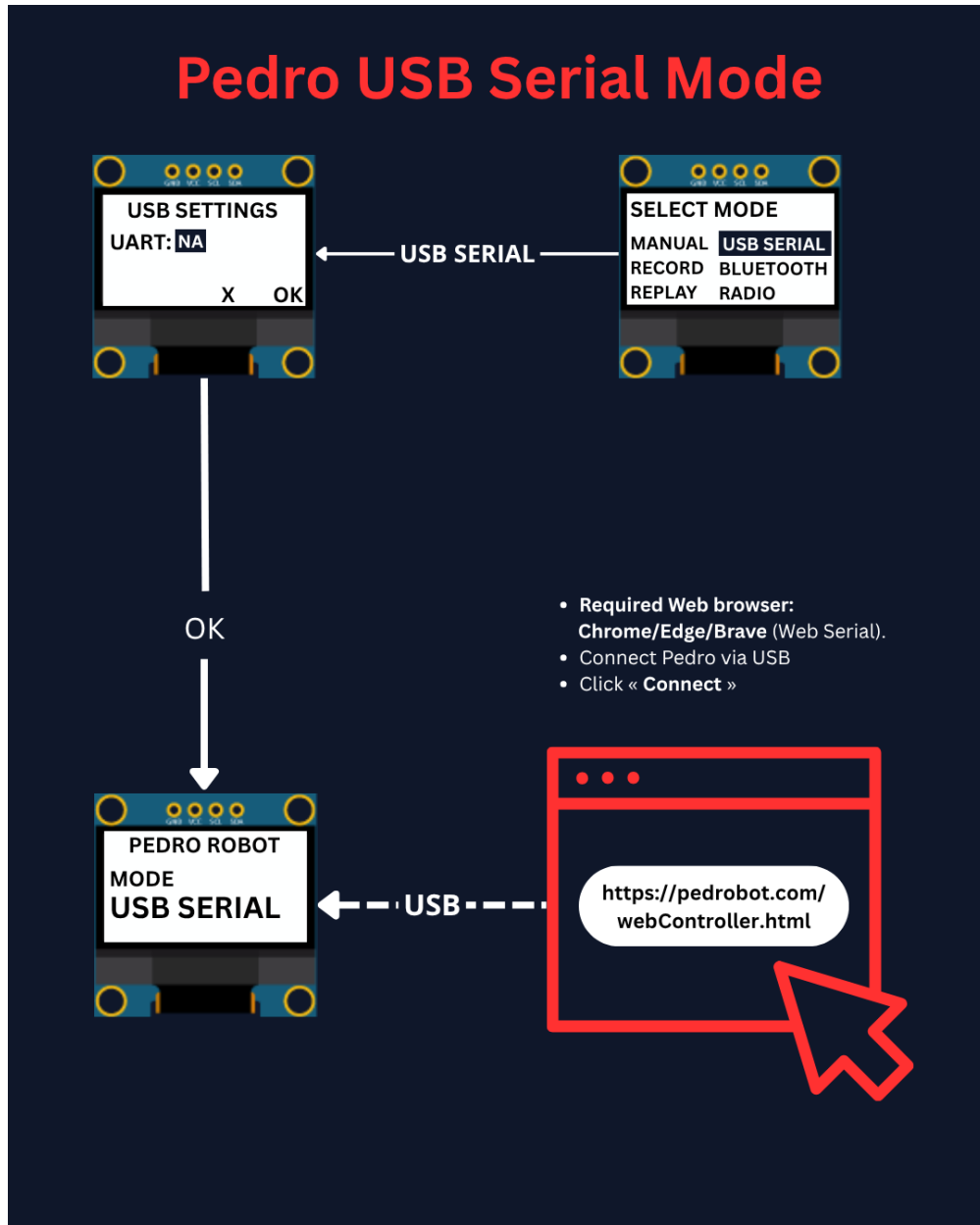


How USB Serial Communication Works





USB Mode Configuration Overview



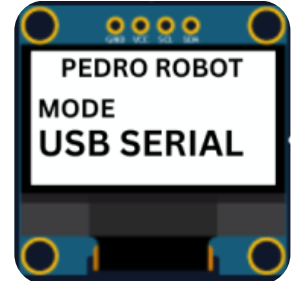
What would happen if the USB cable were disconnected while the robot is moving?
Why does the computer send digital commands instead of directly controlling the servo motor?
Which component of Pedro interprets the USB commands before moving the servos?



Configure Pedro in USB Serial Mode

STEP 1:

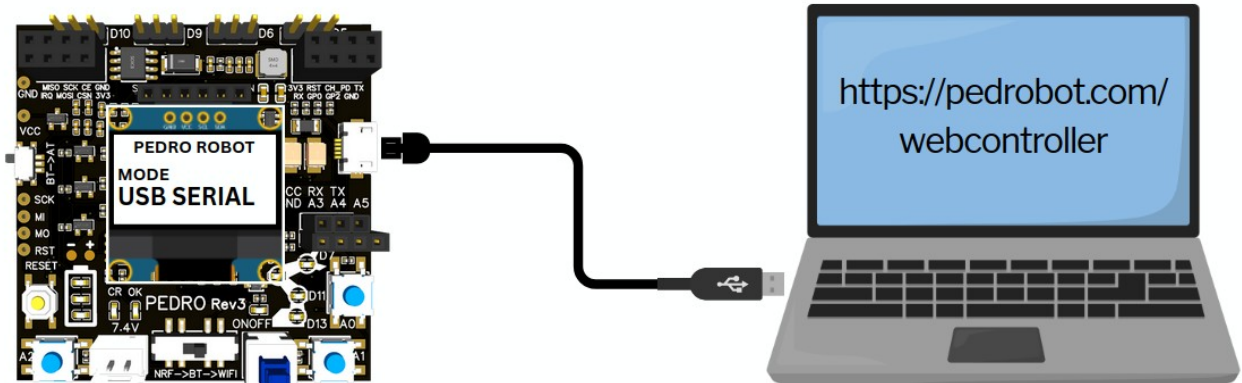
1. Power ON Pedro
2. Enter to the **Select Mode** menu (hold **A0** for 4 seconds)
3. Select **USB SERIAL MODE** mode
4. Press **A0** to confirm
5. The OLED screen should display:



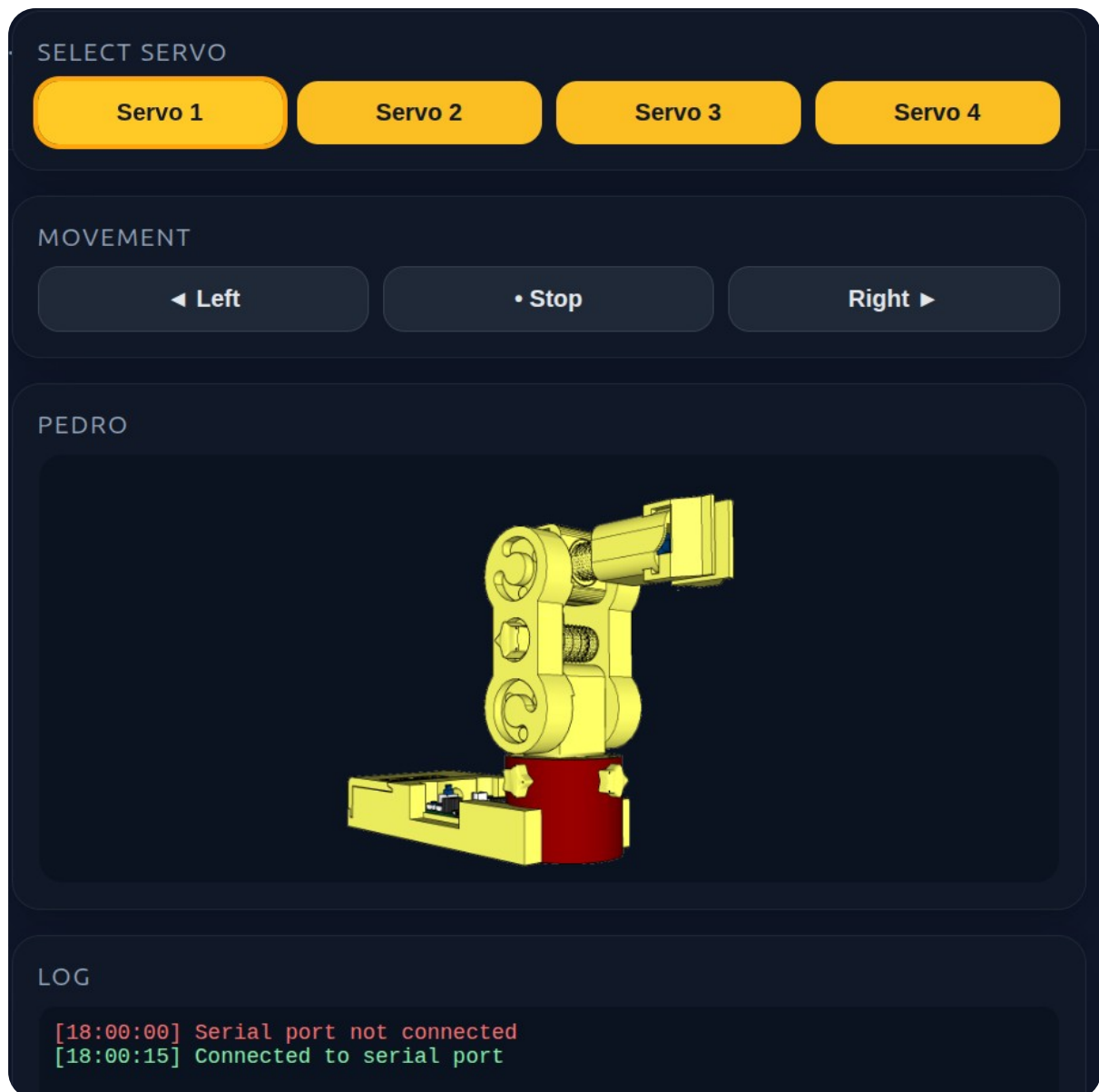
✓ Pedro is now ready to communicate with the computer.

STEP 2:

1. Connect Pedro to your computer using a **USB cable**.
2. Open the **Pedro Web Controller**.
<https://www.pedrobot.com/webController.html>
3. Click **Connect Pedro**.
4. Select Pedro's **serial port**.
5. Click Servo Base to select the servo. If the D13 (Servo Base) LED turns ON, Pedro is successfully connected to the computer.
6. Check that the **D13 LED** turns **ON**.
7. Click **Left** to move the servo left.
8. Click **Right** to move the servo right.
9. Click **Stop** to stop the servo.
10. Try controlling Pedro using the different buttons in the Web Controller.

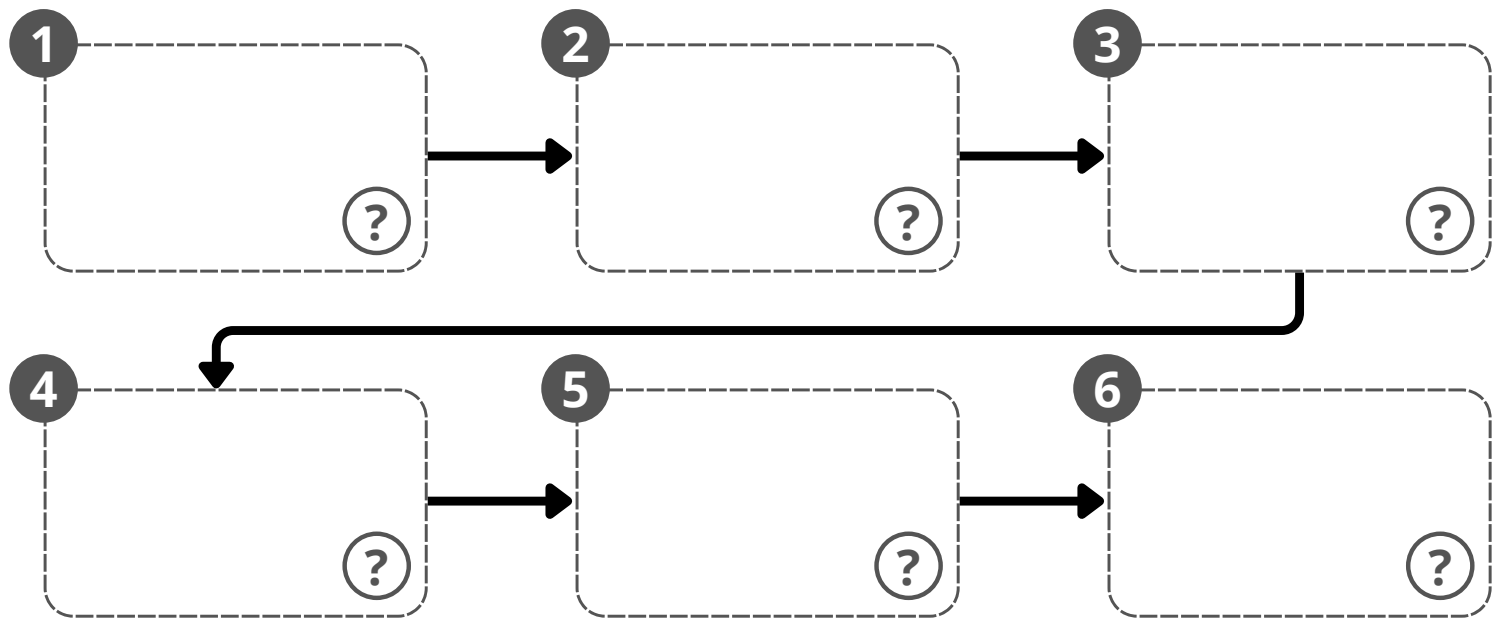


How can you verify that Pedro is correctly connected to your computer?
What is the purpose of selecting a **Servo Base** before sending movement commands?
What would happen if you selected the wrong serial port?



USB Serial Communication Chain

Write the correct letters in boxes 1 to 6 to complete the USB serial communication chain.



A Servo Moves & LED D13 Turns ON

The command becomes real movement and light

B Pedro Web Controller opens in the browser

The Pedro robot is connected via USB port

C ATmega32u4 reads & decodes the data

Figures out which servo and which direction

D Microcontroller generates a PWM signal

Send the electrical pulse to chosen servo

E A USB command is sent through the cable

Contains: selected servo + direction (left/right)

F You click a button (Servo Base + Left)

The student chooses which servo and movement

Once your chain is complete, answer in your own words:



- What would happen at step 3 if the USB cable was unplugged?
- Which step would change if Pedro used Radio NRF24L01 instead of USB?



Quiz - Check Your Understanding

1. What type of communication is used in USB Serial Mode?

- ☐ Hydraulic communication
- ☐ Wireless communication
- ☐ Serial communication
- ☐ Optical communication

2. What is used to connect Pedro to the computer?

- ☐ HDMI cable
- ☐ Ethernet cable
- ☐ USB cable
- ☐ Audio cable

3. What does the web controller allow students to do?

- ☐ Charge the battery
- ☐ Control Pedro from the computer
- ☐ Print 3D parts
- ☐ Edit videos

4. Which browsers are compatible with Pedro's web controller?

- ☐ Chrome, Edge, Brave
- ☐ Safari only
- ☐ Firefox only
- ☐ Internet Explorer

5. What happens when the "Left" button is clicked?

- ☐ The battery charges
- ☐ The OLED turns OFF
- ☐ A serial command moves the servo left
- ☐ The robot resets

6. What does `Serial.read()` do in Arduino?

- ☐ Charges the battery
- ☐ Reads incoming serial data
- ☐ Rotates the servo directly
- ☐ Prints text on the OLED

7. Why is serial communication useful?

- ☐ It changes robot colors
- ☐ It allows reliable data transfer
- ☐ It increases battery size
- ☐ It prints documents

8. What must students select before clicking “Connect Pedro”?

- ☐ WiFi password
- ☐ Camera input
- ☐ Correct serial port
- ☐ OLED brightness

9. Which component receives USB commands inside Pedro?

- ☐ Servo motor
- ☐ OLED display
- ☐ ATmega32U4 microcontroller
- ☐ Battery

10. What engineering concept does USB Serial Mode demonstrate?

- ☐ Human–computer–robot interaction
- ☐ Solar energy conversion
- ☐ Mechanical printing
- ☐ Wireless charging

Congratulations Mission Completed!

You have unlocked: 🏆 the Pedro USB Mode badge
Don't forget to stick your Pedro Badge into your Pedro Passport before starting the next lesson.

